

Department of Telecommunications
Telecommunication Engineering Centre
(IoT Division)

F.No.-1-3/2022-IoT/TEC

Dated 27.12.2022

Subject: Minutes of virtual meeting of National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20 on “Internet of Things (IoT) and Smart Cities & Communities (SC&C)” for the Study Period 2022-24, held on 20th December 2022, 2:30 PM.

The sixth virtual meeting of the National Working Group 20 (NWG-20) corresponding to ITU-T Study Group 20 was held on 20th December 2022 at 2:30 PM through remote participation using C-DOT VC platform. The list of participants is available as *Annexure-1*.

1. Mr. Sushil Kumar, DDG (IoT) welcomed all the participants in the meeting. While briefing about the agenda of the meeting, he mentioned that –

a. This meeting is to finalize the contributions for the next ITU-T SG-20 meeting, scheduled in Jan-Feb 2023.

b. It is worth mentioning that International Telecommunication Union (ITU) has posted following five Technical Reports released by Telecommunication Engineering Centre (TEC) in M2M/ IoT domain in 2021-22, on its website, recognizing them as insightful resource for the global community (<https://www.itu.int/en/ITU-T/climatechange/resources/Pages/Frontier-technologies.aspx#internetofthings>) -

- i. Framework of National Trust Centre for M2M/IoT Devices and Applications
- ii. IoT/ ICT Standards for Smart Cities
- iii. Emerging Communication Technologies & Use Cases in IoT Domain
- iv. Code of Practice for Securing Consumer Internet of Things (IoT)
- v. IoT/ICT Enablement in Smart Village and Agriculture

TEC has released eighteen Technical Reports so far in M2M/ IoT domain covering various verticals, Communication technologies and Security (<https://tec.gov.in/M2M-IoT-technical-reports>).

- c. During the study period 2017-2020, two documents- **ITU-T Recommendation Y Suppl.53 (12/2018)** on IoT Use cases (having five IoT use cases from India and one from Egypt) and **ITU-T Recommendation Y Suppl.56 (12/2019)** on Smart City Use cases (having smart city use cases from Japan, Korea, UK and India), are having significant contributions from India.
 - d. Recently, Q2/20 rapporteur e-meeting was held on 8-13 December 2022, in which the contribution on work item Y.SRC ***“IoT and ICT Requirements for deployment of smart services in rural communities”*** was discussed in detail, its major content has been finalized and target for its consent has been set by rapporteur in the upcoming ITU-T SG-20 meeting, Jan-Feb 2023.
 - e. Next meeting of ITU-T SG-20 FAO Focus Group on ***Artificial Intelligence and IoT for Digital Agriculture (FG-AI4A)*** will be held in virtual mode on 20th January 2023.
 - f. Output drafts of few ongoing work items in SG-20 along with minutes of the last NWG-20 meeting dated 9th Nov 2022 were circulated to all the members and contributions were invited.
 - g. Contribution on the ongoing work item Y.IIoT-infra-SM-fr ***“Requirements and framework of Industrial IoT (IIoT) infrastructure for smart manufacturing”*** and proposal for creating a new work item on Y.RemotEd ***“Requirements for connectivity, bandwidth, privacy and security infrastructure for digital/ online education in remote smart campuses”*** have been received from Mr. Prateep Misra, TCS and Mr. Sharad Arora, Mashmari respectively, for discussion in this meeting.
2. DDG (IoT) presented following ongoing work items (few of them already circulated to the members)-
- a. Y.Sup.SmartAgri-usecases (Use cases of IoT based smart agriculture)

- b. Y.IIoT-infra-SM-fr (Requirements and framework of Industrial IoT (IIoT) infrastructure for smart manufacturing)
- c. Y. Smart-education (Requirements and Reference Architecture of Smart Education)
- d. Y.IoT-SmartBuilding (Common requirements and capabilities of smart buildings from the IoT perspective)
- e. Y.IoT-Smartcity-Risk (Reference framework of cybersecurity risk management of IoT ecosystems in smart cities)

He requested members to send their contributions on these work items of their interest, on priority for discussion in the next NWG-20 meeting, as the last date for submission of contributions in the next ITU-T SG-20 meeting, Jan-Feb 2023, is 17th January 2023.

3. Dr. Gayatri Shakya, JSS Noida asked that can we have some use cases for the solution of problem related with burning of 'Parali', which is one of the major contributor for increasing air pollution these days. DDG (IoT) suggested her to fit-in the proposed solution in the connected agriculture machinery use case in the document Y.Sup.SmartAgri-usecases.

4. Mr. Sharad Arora, Mashmari presented his proposal for creation of new work item on Y.RemotEd ***"Requirements for connectivity, bandwidth, privacy and security infrastructure for digital/ online education in remote smart campuses"***.

DDG (IoT) mentioned that we may go through Y. Smart-education once again and see if this contribution can be integrated in that work item and we may also work on writing justification for this new work item.

Ms. Yashashri Shukla, USOF volunteered to contribute in this new work item.

5. Mr. Prateep Misra, TCS presented his contribution on work item Y.IIoT-infra-SM-fr ***"Requirements and framework of Industrial IoT (IIoT) infrastructure for smart manufacturing"***.

DDG (IoT) mentioned that we may further work on this document along with Mr. Pranav Singh, Idemia, before submission in SG-20 meeting.

6. DDG (IoT) summarized by mentioning that members may send their contributions on the on-going work items (as discussed in the meeting) or any other work item of their interest, on priority by **2nd January 2023** so that same may be discussed in the next NWG-20 meeting on **5th Jan 2023** for further submission in upcoming ITU-T SG-20 meeting, Jan-Feb 2023.

In the end, DDG (IoT) thanked all the participants for attending the meeting.

Way Forwards:

1. Next meeting of NWG-20 will be held on **5th January 2023** at **2:30 PM**.
2. A webinar on *Intelligent Transport System - Technology & Standards* is being organized by TEC on 10th Jan 2023 from 10 AM onwards. All the members are requested to attend the webinar. Link will be shared separately.

This is issued with the consent of DDG (IoT).

Encl: As above

Hindi version shall follow.


(Namrata Singh)
ADG (IoT)
TEC, New Delhi

Annexure-1

S. No.	Name	Organisation
1.	Sushil Kumar	TEC
2.	Ms. Namrata Singh	TEC
3.	Shekhar Singh	TEC
4.	Ms. Jyoti Sengar	TEC
5.	Dr. Bala Murugan MS	VIT Chennai
6.	Harish Kumar	Punjab University
7.	Alok Mittal	STMicroelectronics
8.	Ajay Mittal	TCS
9.	M. Rajith Ali	SIA-India
10.	Pranav Singh	Idemia
11.	Prateep Misra	TCS
12.	Sharad Arora	Mashmari
13.	Dr. Gayatri Sakya	JSS, Noida
14.	Dr. Vydeki D	VIT Chennai
15.	Parag Kulkarni	JohnDeere
16.	Ms. Radhika	NCCS
17.	Ms. Yashashri Shukla	USOF
18.	Vikash	Sensorise